

Laboratory work 3.

Task1. Generate and insert 200 random rows in your airport database.

```
1 INSERT INTO airline (airline_id, airline_code, name, country, created_at, updated_at)
2 SELECT
3     i + 100,
4     'AC' || i,
5     'Airline_' || i,
6     (ARRAY['Kazakhstan', 'France', 'Brazil', 'Poland', 'Turkey'])[i % 5 + 1],
7     CURRENT_TIMESTAMP - (i || ' days')::INTERVAL,
8     CURRENT_TIMESTAMP
9 FROM generate_series(1, 200) AS i
10 RETURNING *;
```

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	airline_id [PK] integer	airline_code character varying (10)	name character varying (100)	country character varying (100)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	AC1	Airline_1	France	2026-09-28 08:06:07.688329	2026-09-29 08:06:07.688329
2	102	AC2	Airline_2	Brazil	2026-09-27 08:06:07.688329	2026-09-29 08:06:07.688329
3	103	AC3	Airline_3	Poland	2026-09-26 08:06:07.688329	2026-09-29 08:06:07.688329
4	104	AC4	Airline_4	Turkey	2026-09-25 08:06:07.688329	2026-09-29 08:06:07.688329
5	105	AC5	Airline_5	Kazakhstan	2026-09-24 08:06:07.688329	2026-09-29 08:06:07.688329
6	106	AC6	Airline_6	France	2026-09-23 08:06:07.688329	2026-09-29 08:06:07.688329
7	107	AC7	Airline_7	Brazil	2026-09-22 08:06:07.688329	2026-09-29 08:06:07.688329
8	108	AC8	Airline_8	Poland	2026-09-21 08:06:07.688329	2026-09-29 08:06:07.688329
9	109	AC9	Ai		2026-09-20 08:06:07.688329	2026-09-29 08:06:07.688329

Total rows: 200 Query complete 00:00:00 1...

"||" is a method of concatenation(AC1,AC2...). (ARRAY['Kazakhstan', 'France', 'Brazil', 'Poland', 'Turkey'])[i % 5 + 1] AS country : Dynamically distributes countries across rows using modular arithmetic, i % 5 returns a remainder between 0 and 4. Adding 1 adjusts the value to match PostgreSQL's 1-based array indexing (1 to 5), cycling evenly through all listed countries.

Task2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.

```
1 INSERT INTO airline (airline_id, airline_code, name, country, created_at, updated_at)
2 VALUES (
3     301,
4     'KZA',
5     'KazAir',
6     'Kazakhstan',
7     CURRENT_TIMESTAMP,
8     CURRENT_TIMESTAMP
9 )
10 RETURNING *;
```

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	airline_id [PK] integer	airline_code character varying (10)	name character varying (100)	country character varying (100)	created_at timestamp without time zone	updated_at timestamp without time zone
1	301	KZA	KazAir	Kazakhstan	2026-09-29 08:21:34.647201	2026-09-29 08:21:34.647201

Task3. Update the airline country "KazAir" to "Turkey".

```
1 UPDATE airline SET country = 'Turkey' WHERE name = 'KazAir' RETURNING *;
```

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	airline_id [PK] integer	airline_code character varying (10)	name character varying (100)	country character varying (100)	created_at timestamp without time zone	updated_at timestamp without time zone
1	301	KZA	KazAir	Turkey	2026-09-29 08:21:34.647201	2026-09-29 08:21:34.647201

Task4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".

```
1 INSERT INTO airline (airline_id, airline_code, name, country, created_at, updated_at)
2 VALUES
3     (302, 'AC302', 'AirEasy', 'France', NOW(), NOW()),
4     (303, 'AC303', 'FlyHigh', 'Brazil', NOW(), NOW()),
5     (304, 'AC304', 'FlyFly', 'Poland', NOW(), NOW())
6 RETURNING *;
```

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	airline_id [PK] integer	airline_code character varying (10)	name character varying (100)	country character varying (100)	created_at timestamp without time zone	updated_at timestamp without time zone
1	302	AC302	AirEasy	France	2026-09-29 08:33:17.267395	2026-09-29 08:33:17.267395
2	303	AC303	FlyHigh	Brazil	2026-09-29 08:33:17.267395	2026-09-29 08:33:17.267395
3	304	AC304	FlyFly	Poland	2026-09-29 08:33:17.267395	2026-09-29 08:33:17.267395

Task 5. Delete all flights whose arrival in 2024 year.

```
1 DELETE FROM flight
2 WHERE sch_arrival_time BETWEEN '2024-01-01 00:00:00' AND '2024-12-31 23:59:59'
3 RETURNING *;
```

Task 6. Increase the price of all tickets in booking table for flights by 15%.

```
1 UPDATE booking SET price = price * 1.15 RETURNING *;
```

Task 7. Delete all tickets where price is less than 10000.

```
1 DELETE FROM booking WHERE price < 10000 RETURNING *;
```

Task8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).

```
1 UPDATE airline SET airline_code = 'UNK' WHERE airline_code IS NULL
2 RETURNING *;
```

Task9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.

```
1 DELETE FROM baggage_check WHERE created_at < '2023-06-01' AND status = 'Not checked'
2 RETURNING *;
```

Task10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.

```
1 DELETE FROM airport WHERE state IS NULL AND city IN ('Mlawe', 'Kepuh')
2 RETURNING *;
```

Task11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.

```
1 INSERT INTO baggage_check (baggage_check_id, check_result, status, created_at, updated_at, passenger_id)
2 VALUES (9, 'Pending', 'Not checked', NOW(), NOW(), 1)
3 RETURNING baggage_check_id, created_at;
```

Task12. Convert all country names in the airline table to uppercase to ensure data consistency.

```
1 UPDATE airline SET country = UPPER(country) RETURNING *;
```

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	airline_id [PK] integer	airline_code character varying (10)	name character varying (100)	country character varying (100)	created_at timestamp without time zone	updated_at timestamp without time zone
1	101	AC1	Airline_1	FRANCE	2026-09-28 08:06:07.688329	2026-09-29 08:06:07.688329
2	102	AC2	Airline_2	BRAZIL	2026-09-27 08:06:07.688329	2026-09-29 08:06:07.688329
3	103	AC3	Airline_3	POLAND	2026-09-26 08:06:07.688329	2026-09-29 08:06:07.688329
4	104	AC4	Airline_4	TURKEY	2026-09-25 08:06:07.688329	2026-09-29 08:06:07.688329
5	105	AC5	Airline_5	KAZAKHSTAN	2026-09-24 08:06:07.688329	2026-09-29 08:06:07.688329
6	106	AC6	Airline_6	FRANCE	2026-09-23 08:06:07.688329	2026-09-29 08:06:07.688329
7	107	AC7	Airline_7	BRAZIL	2026-09-22 08:06:07.688329	2026-09-29 08:06:07.688329

```
1 UPDATE airline
2 SET name = 'Global Airways',
3     country = 'United Kingdom',
4     updated_at = CURRENT_DATE
5 WHERE airline_id = 105
6 RETURNING *;
```

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	airline_id [PK] integer	airline_code character varying (10)	name character varying (100)	country character varying (100)	created_at timestamp without time zone	updated_at timestamp without time zone
1	105	AC5	Global Airways	United Kingdom	2026-09-24 08:06:07.688329	2026-09-29 00:00:00

```
1 UPDATE airport SET state = 'Capital District' WHERE city IN ('Astana', 'London', 'Tokyo')
2 RETURNING *;
```

```
1 UPDATE baggage_check SET status = 'Checked'
2 WHERE status = 'Not checked' AND created_at >= '2024-03-01' AND created_at < '2024-04-01'
3 RETURNING *;
```